The Spiral and the Wave

Scale-Invariant Geometry for Broadband Electromagnetic and Acoustic Absorption

An Invitation to Collaborate

"The logarithmic spiral is the only curve that is scale-invariant. This geometric property is critical because it ensures broadband absorption... As the wave frequency changes, the active part of the spiral simply shifts along the curve, but the absorption mechanism remains identical."

— Wang, Hou & Chan, Scientific Reports (2019)

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Executive Summary

The Problem

Current stealth technology—whether electromagnetic or acoustic—relies on platform-specific geometry, frequency-tuned coatings, or complex active cancellation systems. These approaches are expensive, narrowband, and tightly coupled to vehicle design. Broadband absorption typically requires stacking multiple resonators tuned to different frequencies, increasing thickness, complexity, and fragility.

The Core Insight

A 2019 paper by Wang, Hou, and Chan demonstrated that logarithmic spiral metasurfaces achieve greater than 95% electromagnetic absorption across a six-octave bandwidth (6–37 GHz) using a single geometric motif. The mechanism exploits the unique scale invariance of the logarithmic spiral: as frequency changes, the active absorption region shifts along the curve, but the physics remains identical. The authors explicitly noted this principle "may also be applied to the absorption of other classical waves such as sound."

The Question

Can this scale-invariant absorption principle transfer from electromagnetic waves to acoustic waves—specifically, sound in water? The mathematical structures governing these domains share deep similarities. If the transfer works, it would decouple stealth from platform design and enable materials-based broadband absorption applicable as coatings to existing structures.

The Ask

This document invites collaboration from experts in metamaterials, acoustic engineering, piezoelectric materials, and wave physics. Key questions: Has the electromagnetic result been independently replicated? Where does the electromagnetic-to-acoustic analogy break down? What experimental validation is required? The author seeks intellectual exchange—not funding—to determine whether this concept merits further investigation.

1. The Observation

In 2019, a team at the Hong Kong University of Science and Technology published a paper in Scientific Reports describing a metasurface composed of logarithmic spiral resonators that could absorb more than 95% of incident microwave energy across an extraordinarily broad frequency range—from 6 GHz to 37 GHz.

This was not achieved through the conventional approach of stacking multiple resonators tuned to different frequencies. Instead, Wang, Hou, and Chan exploited a fundamental property of the logarithmic spiral: it is the only shape that remains unchanged under scaling. Rotate a logarithmic spiral by any angle, and it looks identical to scaling it by some factor. This scale invariance, combined with Fabry-Pérot-type resonances, created something simple and elegant—a single geometric motif that naturally handles the entire microwave spectrum.

The thickness was just 5 mm—approaching the theoretical Rozanov limit for optimal absorbers.

2. The Central Question

If the logarithmic spiral can trap and absorb electromagnetic waves across a six-octave bandwidth using nothing but geometry and lossy dielectric, can the same principle be applied to other wave phenomena?

Specifically: acoustic waves. Sound in water.

Maxwell's equations for electromagnetic waves and the equations governing acoustic shear-horizontal modes share a deep mathematical structure. Both are wave equations. Both respect certain symmetries. The question is whether the scale-invariant absorption demonstrated by Wang et al. in the electromagnetic domain might transfer—partially or wholly—to the acoustic domain.

3. Strategic Significance

The current state of stealth technology—whether radar or sonar—relies heavily on platform geometry (shaping), frequency-specific coatings (tuned absorbers), or active cancellation systems that are expensive, complex, and often fragile.

A materials-based approach that achieves broadband absorption through pure geometry would be qualitatively different. It would decouple stealth from platform design. It could potentially be applied as a coating to existing structures. And if the geometry is self-similar, it might maintain performance across a wide range of operating conditions.

3.1 The Air-Water Impedance Asymmetry

There is an intriguing asymmetry between air and water applications. The impedance mismatch between piezoelectric materials (which could actuate the spirals) and air is roughly 72,000:1. In water, that ratio drops to approximately 20:1, or even 7:1 with the right composite materials. This suggests that if the principle works at all, it might work better underwater—where acoustic stealth is strategically critical and current passive tile technologies have well-known depth limitations.

These are speculations. The physics needs to be tested.

4. What We Know

4.1 The Wang et al. Result

The 2019 paper demonstrated that a metasurface of logarithmic spiral resonators (defined by $r = \alpha e^{\beta\theta}$), backed by a metal surface and filled with lossy dielectric material, can achieve greater than 95% absorption from 6–37 GHz at normal incidence. The critical insight was that the spiral acts like a "space-coiled" tapered parallel-plate waveguide, creating Fabry-Pérot resonances whose frequencies shift along the curve as wavelength changes—but the absorption mechanism remains the same.

In contrast, an Archimedean spiral ($r = r_0 + \rho\theta$), which lacks scale invariance, only achieved strong absorption at discrete frequencies. The difference is stark and directly attributable to geometry.

The authors noted that their approach "may also be applied to the absorption of other classical waves such as sound."

4.2 The State of Underwater Acoustic Metamaterials

Active research exists on underwater acoustic metamaterials, including spiral-based designs. Recent work includes spiral resonators embedded in polyurethane matrices, space-coiled water channels with rubber coatings, and various local-resonance structures. These approaches have achieved varying degrees of success at low frequencies, though typically with narrower bandwidths than the electromagnetic case.

What has not been found is a direct experimental test of the specific claim: that a logarithmic spiral geometry provides broadband acoustic absorption because of scale invariance—the same mechanism Wang et al. demonstrated for microwaves.

This gap is either an oversight in the literature search, a result that exists but has not been published, or an experiment that has not been done.

5. What We Do Not Know

The following uncertainties are genuine unknowns that would need to be resolved experimentally:

5.1.1 Independent Replication

It is unknown whether Wang et al. (2019) has been independently replicated. The paper has been cited, built upon, and referenced—but no published independent replication of the core result has been located. This is the single most important unknown. If the electromagnetic result does not replicate, everything downstream collapses.

5.1.2 Physics Transfer to Acoustics

It is unknown how the physics transfers to acoustics. The mathematical analogy between Maxwell's equations and acoustic wave equations is real, but analogies have limits. The dielectric loss mechanisms in the electromagnetic case may not map cleanly to viscous dissipation in the acoustic case. The boundary conditions differ. The material requirements differ.

5.1.3 Practical Fabrication

It is unknown whether practical fabrication is achievable. Wang et al. suggested a "Swiss Roll" manufacturing process—coating metal with wedge-shaped dielectric and rolling it. Whether this translates to underwater acoustic materials, with their different mechanical requirements, is unclear.

5.1.4 Angle-of-Incidence Performance

It is unknown what the angle-of-incidence performance for acoustics would be. The electromagnetic results showed good absorption within ±30° of normal incidence. Acoustic requirements may differ significantly.

6. The Core Physics Argument

Here is the argument, stripped to its essentials:

1. Broadband absorption is hard because most resonators work at specific frequencies. To absorb broadly, you typically need to stack many resonators—each tuned to a different band—which increases thickness, complexity, and fragility.

2. The logarithmic spiral is the only curve that is scale-invariant. Scaling it is equivalent to rotating it. This means a single spiral "contains" resonators at all scales simultaneously.

3. Wang et al. showed that this geometric property translates directly to broadband electromagnetic absorption. As frequency changes, the "active region" of the spiral shifts—but the absorption mechanism persists.

4. The 2D Maxwell equations for TM modes and the equations for shear-horizontal acoustic waves are mathematically equivalent. This suggests (but does not prove) that the same geometry might achieve analogous results for sound.

5. If true, this would provide a route to broadband acoustic absorption using a single geometric motif—potentially transforming underwater stealth from a platform-specific engineering problem to a materials-science problem.

The logic is sound. The physics is plausible. The experimental validation is missing.

7. An Invitation

This is not a funding request. This is not a programme proposal. This is a question and an invitation to think about it together.

If you have expertise in metamaterials, acoustic engineering, piezoelectric materials, submarine acoustics, or simply geometry and wave physics—the following perspectives would be valuable:

- **Does the Wang et al. result replicate?** Has anyone in your laboratory or network attempted this? What did you find?
- **Is the EM-to-acoustic analogy valid?** Where does the mathematical equivalence break down in practice? What are the failure modes?
- **What is missing?** Is there a body of literature not found? A fundamental objection not considered? A reason this has been tried and abandoned?
- **Would you be interested in exploring this together?** Not as a formal collaboration necessarily, but as an intellectual exchange. A few emails. A conversation. A shared document.

This question is worth answering. If the answer is "no, it does not work," that is valuable—it prevents wasted effort. If the answer is "yes, and here is why," that could be consequential.

8. What the Author Brings

Considerable time has been invested developing the conceptual framework for this idea—including extensive technical documentation exploring failure modes, governance challenges, and experimental validation requirements. The literature has been mapped across electromagnetic metamaterials, piezoelectric materials science, radar systems, and acoustic wave physics.

The work has been subjected to rigorous adversarial review—deliberately searching for logical faults, circular reasoning, and structural weaknesses. There is awareness that the enthusiasm of the inventor is a liability, not an asset, when evaluating novel concepts.

What is lacking is the experimental capability to answer the central question. That requires facilities, expertise, and time that cannot be provided alone.

Hence this invitation.

9. A Final Note

The logarithmic spiral appears throughout nature—in nautilus shells, galaxy arms, hurricane bands, and the flight paths of hawks. It was called the "spira mirabilis" by Jacob Bernoulli, who was so enamoured with its properties that he requested it be engraved on his tombstone.

There is something poetic in the idea that a shape so fundamental to the natural world might also hold the key to making objects invisible to electromagnetic and acoustic detection. Whether that poetry corresponds to physical reality—that is what needs to be found out.

References

Wang, S., Hou, B. & Chan, C.T. "Broadband Microwave Absorption by Logarithmic Spiral Metasurface." Scientific Reports 9, 14078 (2019). https://doi.org/10.1038/s41598-019-50603-4

This paper is open access and freely available. Anyone interested in this concept is encouraged to read it directly.

Appendix A: Glossary

Term	Definition
Archimedean Spiral	A spiral defined by $r = r_0 + \rho\theta$; lacks scale invariance
Fabry-Pérot Resonance	Resonance phenomenon occurring when waves reflect between parallel surfaces, creating standing wave patterns
GHz	Gigahertz; one billion cycles per second
Impedance Mismatch	Difference in wave propagation characteristics between two media, affecting energy transfer
Logarithmic Spiral	A spiral defined by $r = \alpha e\hat{}\beta\theta$; uniquely scale-invariant
LSR	Logarithmic Spiral Resonator
Metasurface	An artificially structured surface designed to manipulate electromagnetic or acoustic waves
Rozanov Limit	Theoretical minimum thickness for an optimal broadband absorber
Scale Invariance	Property where a shape appears identical regardless of magnification; rotating equals scaling
TM Mode	Transverse Magnetic mode; electromagnetic wave configuration where the magnetic field is perpendicular to the direction of propagation

Appendix B: Document Metadata

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